

Traffic Signal Violation Detection at Simpang Wirosaban Using YOLOv11, HSV Thresholding, and SORT

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Abstract—This is a placeholder abstract. Replace this text with a concise summary of the research background, method, main findings, and conclusion.

I. INTRODUCTION

A. Background

Traffic signal violations, particularly red light running, constitute one of the most prevalent and dangerous forms of traffic misconduct at urban intersections. [1] In Indonesia, the problem has reached a critical scale. Data from the National Criminal Information Center of the Indonesian National Police recorded 148,798 traffic accidents throughout 2023, of which 94.63% were attributed to human error [2]. The Director of Traffic Accident Investigation of the Indonesian National Police further stated that every accident is preceded by a violation, underscoring the direct causal relationship between traffic infractions and collision events [2]. Simpang Wirosaban, a signalized intersection in Yogyakarta, represents a typical urban intersection in Indonesia where such violations occur at high frequency, particularly during nighttime hours when traffic enforcement is minimal. The absence of automated monitoring at such locations necessitates a technology-driven approach to continuous, objective violation detection.

Conventional methods of traffic enforcement rely on human officers stationed at intersections or on radar-based sensors embedded in road infrastructure. These approaches are costly to deploy and maintain, limited in spatial coverage, and susceptible to human fatigue and inconsistency [3]. The proliferation of closed-circuit television (CCTV) infrastructure at intersections across Indonesian cities presents an opportunity to repurpose existing hardware for automated enforcement

without additional installation costs. Computer vision techniques applied to CCTV footage have demonstrated considerable promise in this domain. Ong et al. [3] proposed a YOLO-based system targeting red light crossing and double-line overtaking violations from CCTV footage, establishing a foundational framework for vision-based violation detection. Similarly, Tomas et al. [4] demonstrated that YOLOv5 combined with Faster R-CNN achieved high accuracy in detecting red light and wrong-way violations at intersections, reinforcing the viability of deep learning pipelines for this task.

Vehicle detection in video-based systems has been significantly advanced by the YOLO family of single-stage object detectors. Since its introduction, YOLO has undergone continuous architectural refinement, with each iteration improving detection accuracy and inference speed. YOLO11, released by Ultralytics in September 2024 at YOLO Vision 2024, introduced C3K2 blocks, an enhanced Spatial Pyramid Pooling Fast (SPPF) module, and a C2PSA partial spatial attention mechanism [5]. The YOLO11m variant achieves higher mean average precision on the COCO benchmark with 22% fewer parameters than its YOLOv8m predecessor, making it well-suited for real-time deployment on constrained hardware [5]. For multi-object tracking across video frames, this study employs the Simple Online and Realtime Tracking (SORT) algorithm proposed by Bewley et al. [6], which combines the Kalman filter for state estimation with the Hungarian algorithm for frame-to-frame data association. Despite its computational simplicity, SORT achieves tracking performance comparable to significantly more complex methods while operating at over 260 Hz, satisfying the latency requirements of real-time traffic monitoring.

A persistent challenge in nighttime traffic surveillance is the degradation of image quality caused by uneven illumination, vehicle headlight glare, and sensor noise, all of which reduce object detection confidence. Lashkov et al. [7] demonstrated that Contrast Limited Adaptive Histogram Equalization (CLAHE) applied as a preprocessing step substantially improves vehicle detection performance in nighttime conditions. Unlike standard histogram equalization, CLAHE operates on localized image tiles and enforces a contrast limit to prevent noise amplification, making it particularly effective for scenes with spatially non-uniform lighting such as signalized intersections at night [8], [9]. In this study, CLAHE is applied exclusively to the luminance channel of the LAB color space prior to YOLO11 inference, preserving the chrominance information required by the HSV-based traffic light detector. This decoupled enhancement strategy ensures that preprocessing improvements to vehicle detection do not introduce artifacts into the color-sensitive components of the pipeline.

B. Objective

The primary objectives of this study include:

- To develop a real-time traffic violation detection system capable of identifying vehicles that cross the stop line during a red light signal using CCTV footage.
- To implement vehicle detection and multi-object tracking using YOLOv11 and SORT algorithm to accurately localize and track vehicles within a defined detection zone.
- To design a rule-based violation logic that determines infractions based on vehicle trajectory relative to the stop line and the current traffic light state.
- To evaluate the effectiveness of image enhancement techniques, including CLAHE, Gamma Correction.

C. Scope

The scope of the study includes:

- The study only can be implemented at the Simpang Wirosaban, Yogyakarta, using footage from a fixed CCTV camera.
- The system is optimized for nighttime conditions, where traffic light colors are more distinguishable.
- Detection performance may decrease under rainy weather, poor visibility, or varying lighting conditions.

II. LITERATURE REVIEW

A. Related Work

B. HSV (Hue, Saturation, Value) Color Model

In the field of computer vision, selecting an appropriate color space is critical for the accuracy of object detection under varying environmental conditions. The HSV (Hue, Saturation, Value) color model is a cylindrical representation of colors that more closely aligns with human visual perception compared to the standard RGB model. While RGB defines colors as a combination of primary light sources, it is highly sensitive to changes in brightness, shadows, and external lighting. The

HSV model decouples color information from intensity, using three distinct components:

- Hue (H): Represents the dominant wavelength or the actual "color" (e.g., Red, Yellow, Green) expressed in degrees from 0° to 360° .
- Saturation (S): Indicates the vibrancy or purity of the color, ranging from 0 to 100%.
- Value (V): Represents the brightness or luminance of the color.

C. You Only Look Once 11 (YOLO11)

D. Simple Online and Realtime Tracking (SORT)

E. Contrast Limited Adaptive Histogram Equalization (CLAHE)

F. Gamma Correction

G. Gaussian Blur

III. METHODOLOGY

A. Proposed Approach

This study proposes a computer vision-based approach for detecting traffic signal violations using CCTV footage from a static camera installed at Simpang Wirosaban, Yogyakarta. The system is designed as a multi-stage pipeline (Figure 1) consisting of image enhancement, vehicle detection, multi-object tracking, traffic light recognition, and violation analysis. To reduce computational overhead and eliminate detections from irrelevant regions such as sidewalks and surrounding infrastructure, a polygon-based Region of Interest (ROI) is defined to constrain the analysis strictly to the active roadway area.

Prior to vehicle detection, each frame undergoes a three-stage image enhancement pipeline to improve visibility under nighttime and uneven lighting conditions. Gamma correction is first applied to brighten underexposed regions by adjusting pixel intensity according to a tunable gamma value ($\gamma = 1.5$). The gamma-corrected frame is then passed to Contrast Limited Adaptive Histogram Equalization (CLAHE), applied exclusively to the luminance channel of the LAB color space to enhance local contrast without introducing chrominance distortion. Finally, Unsharp Masking is applied to sharpen object boundaries by subtracting a Gaussian-blurred version of the frame from the original, which mitigates the effect of motion blur on moving vehicles. This enhancement pipeline is applied solely to the input of the vehicle detector; the original unmodified frame is retained for all other processing stages to preserve color fidelity.

Within the defined ROI, vehicle detection is performed using YOLO11, which identifies vehicles present inside the polygonal area on each frame. The detected bounding boxes are subsequently processed by the Simple Online and Realtime Tracking (SORT) algorithm, which employs a Kalman filter for state prediction and the Hungarian algorithm for frame-to-frame identity association. This tracking mechanism assigns a persistent unique identifier to each vehicle and maintains

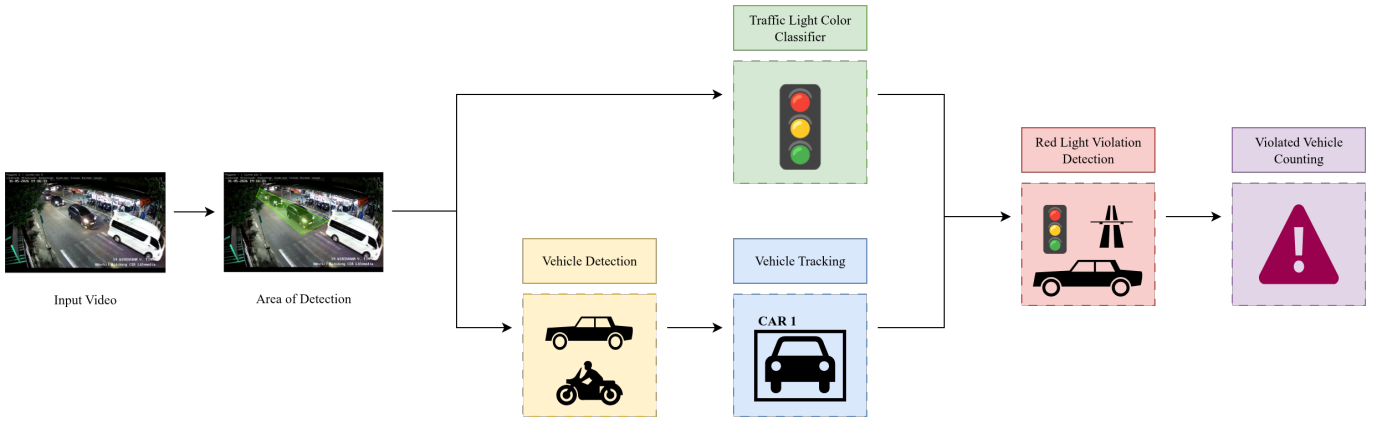


Fig. 1: Diagram flow of traffic signal violation detection at Simpang Wirosaban.

its trajectory across consecutive frames, enabling reliable monitoring of individual vehicle movements over time.

In parallel with vehicle detection, the traffic light state is determined using an HSV color-based thresholding method applied to a dedicated ROI positioned at the traffic light location in the frame. The detected state is used as a conditional input to the violation logic module. A virtual stop line is defined within the ROI to represent the legal stopping boundary at the intersection. Violation detection is triggered when a tracked vehicle, whose trajectory originated from within the ROI, crosses the stop line at the moment the traffic light is in a red state. To prevent duplicate records, each unique vehicle identity is registered as a violation at most once. The system produces output in the form of annotated video frames displaying bounding boxes, vehicle identities, stop line geometry, and a running violation counter, providing a structured and automated means of traffic violation documentation.

B. Dataset

Write the dataset description here.

C. Image Enhancement

Write the preprocessing steps here.

D. Detection and Tracking

Write the detection and tracking method here.

E. Violation Logic

Write the violation decision logic here.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

The experiments were conducted using input videos with a resolution of 1920×1200 pixels to preserve fine-grained details of vehicle motion and traffic light status. All video processing and model inference were executed on a workstation equipped with an NVIDIA Tesla T4 GPU, which provided hardware acceleration for YOLO11 detection and SORT-based tracking during the evaluation stage.

B. Results

Write the main results here.

V. CONCLUSION

A. Conclusion

Write the final conclusion here.

B. Future Work

Write possible future improvements here.

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